INTERNATIONAL JOURNAL OF
RADIATION ONCOLOGY • BIOLOGY • PHYSICS

www.redjournal.org

REVIEW

Postoperative Adjuvant Treatments for Cervical Cancer: A Network Meta-Analysis

Wenrui Huang, PhD,[a,b] Xingzi Fang, MD,[a,b] Xingyan Ou, MD,[a,b] Lei Chen, MD,[a,b] Xiaoxuan Tang, MD,[a,b] and Xuelian Du, MD, PhD[a,b]

[a]*Department of Gynecology, The Fourth Clinical Medical College of Guangzhou University of Chinese Medicine, Shenzhen, Guangdong, China; and* [b]*Department of Gynecology, Shenzhen Traditional Chinese Medicine Hospital, Shenzhen, Guangdong, China*

Received Nov 20, 2024; Revised Jun 24, 2025; Accepted for publication Jul 22, 2025

This study aimed to perform a comprehensive network meta-analysis assessing the efficacy and safety of 5 postoperative adjuvant therapies (PATs) for early-stage cervical cancer, comparing their effects on survival and adverse outcomes to identify the therapy with the most significant survival benefit. We comprehensively searched PubMed, Web of Science, the Cochrane Library, Embase, CNKI, VIP, Wanfang, and CBM for randomized controlled trials on PATs—including concurrent chemoradiation therapy (CCRT), radiation therapy (RT), chemotherapy (CT), sequential chemoradiation therapy (SCRT), and CCRT followed by consolidation chemotherapy (CCRT + CT)—in patients with cervical cancer from inception to October 2024. Study bias was assessed with the Cochrane ROB.2. A Bayesian random-effects network meta-analysis was conducted for indirect comparisons, with τ^2 used to evaluate heterogeneity. Treatment rankings were generated using the surface under the cumulative ranking (SUCRA) curve, and evidence certainty was assessed with the Confidence in Network Meta-Analysis framework. Nineteen randomized controlled trials with a total of 2937 patients met the inclusion criteria. In our network meta-analysis, CCRT + CT showed the most significant benefit for overall survival (hazard ratios [HR], 0.46; 95% CI, 0.28-0.71; SUCRA 95%, high-certainty evidence), followed by SCRT (1.59; 95% CI, 1.04-2.33; SUCRA 65%, high-certainty evidence) and CCRT alone (HR, 0.71; 95% CI, 0.5-0.95; SUCRA 48%, moderate-certainty evidence). For disease-free survival, CCRT + CT again showed a significant advantage over RT (HR, 0.48; 95% CI, 0.23-0.91; SUCRA 88%, low-certainty evidence). In recurrence analysis, CCRT + CT demonstrated the most substantial reduction in relative risk (RR, 0.45; 95% CI, 0.15-0.83; 85%), followed by CCRT (RR, 0.59; 95% CI, 0.24-0.97; SUCRA 68%) and SCRT (RR, 1.17; 95% CI, 1.01-3.28; SUCRA 40%). No statistically significant differences were observed for distant recurrence or metastasis among the therapies; however, SUCRA rankings indicated that CCRT + CT and CCRT might offer some benefit in reducing these outcomes. For adverse events—specifically myelosuppression, proctitis, cystitis, and nausea/vomiting—no significant differences in risk were observed across therapies. However, SUCRA rankings favored RT as the most favorable option for minimizing these events. This study provides a comprehensive assessment of PATs for early-stage cervical cancer. Among the evaluated options, CCRT + CT showed the most promising trend toward improving overall and disease-free survival and reducing recurrence, particularly in patients with high-risk features. However, further high-quality studies are needed to confirm these findings and guide treatment decisions for specific subgroups.

Corresponding author: Xuelian Du, MD, PhD; E-mail: jndxl@hotmail.com

Author Responsible for Statistical Analyses: Dr wenrui huang.

Disclosures: The authors have no financial or proprietary interests in any material discussed in this article. This work was financially supported by the Sanming Project of Medicine in Shenzhen (No. SZZYSM202211006).

Data Sharing Statement: No additional data and materials are available.

Acknowledgments—We thank all the authors of the previous studies who kindly provided additional information and data from their studies for this meta-analysis.

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.ijrobp.2025.07.1440.

Int J Radiation Oncol Biol Phys, Vol. 124, No. 1, pp. 98−108, 2026
0360-3016/$ - see front matter
https://doi.org/10.1016/j.ijrobp.2025.07.1440

Introduction

Cervical cancer is the fourth most common cancer among women worldwide, with a significant mortality rate.[1] Recent estimates from 2022 indicate approximately 662,000 new cases and 349,000 deaths from cervical cancer annually worldwide.[2] The National Comprehensive Cancer Network (NCCN) guidelines recommend radical hysterectomy, bilateral pelvic lymphadenectomy, and optional oophorectomy for stages IA, IB, and some IIA1.[3] However, the need for further postoperative treatment depends on the pathological risk factors identified after surgery.

After surgery, patients with early-stage cervical cancer are commonly categorized into high-risk or intermediate-risk groups based on pathological findings. For those with high-risk features—such as lymph node involvement, parametrial invasion, or positive surgical margins—concurrent chemoradiation therapy (CCRT) with cisplatin is recommended as the standard postoperative adjuvant therapy (PAT), according to the current NCCN guidelines.[3] This approach is supported by evidence from the GOG 109 trial, which showed that CCRT significantly improves both disease-free survival (DFS) and overall survival (OS) in this population.[4] In contrast, patients with intermediate-risk factors—such as tumor size greater than 4 cm, deep stromal invasion, or lymphovascular space invasion—may benefit from postoperative pelvic radiation therapy (RT). The GOG 92 trial demonstrated a reduction in recurrence rates, particularly among those with multiple risk factors.[5,6] However, this study did not show a significant improvement in OS. Importantly, the optimal strategy for managing intermediate-risk patients remains unclear. International guidelines vary, particularly regarding the use of concurrent chemotherapy and whether adjuvant treatment is necessary in cases with a single intermediate-risk factor. For example, the European Society for Medical Oncology guidelines consider surveillance a reasonable option for patients with only one such factor.[7]

Beyond the standard regimen, recent studies have investigated alternative postoperative strategies—such as chemotherapy (CT) alone, sequential chemoradiation therapy (SCRT), or CCRT followed by consolidation chemotherapy (CCRT + CT) —in an effort to enhance survival while minimizing toxicity and improving treatment adherence. However, the effectiveness of these treatments remains debated. Some studies reported similar DFS rates between SCRT and CCRT, whereas others found SCRT to be superior. Additionally, some research suggested that CT provided better OS and DFS than CCRT, yet other studies showed no significant survival differences between CCRT, CT, or RT.[8-12] These inconsistent findings highlight the uncertainty about the efficacy of PATs in early-stage cervical cancer, emphasizing the need for a comprehensive meta-analysis of all published trials.

Despite several existing meta-analyses, no network meta-analysis (NMA) has been conducted. Furthermore, most previous meta-analyses have focused on survival and recurrence/metastasis,[13-15] with limited attention to adverse reactions and influencing factors. Some studies combined survival data using odds ratios or relative risk (RR), potentially overlooking the impact of survival duration,[16] which could introduce bias.[17] Therefore, this NMA integrated survival data using HR and their 95% confidence intervals (CIs) to evaluate the 5 PATs for early-stage cervical cancer. The goal was to identify the treatment with the most significant survival benefit and the least amount of bone marrow suppression.

Methods

This NMA followed the PRISMA 2020 guidelines and the PRISMA-NMA extension.[18,19] The registration details are available in Appendix E1. No ethical approval was required because this study synthesized evidence on published randomized controlled trials (RCTs) that included no confidential data and personal information about the patients. All authors have read and approved the final manuscript.

Search strategy

We conducted an extensive search across 8 databases: CNKI, VIP, Wanfang, CBM, PubMed, Web of Science, Cochrane, and Embase, targeting RCTs on PATs for cervical cancer patients in International Federation of Gynecology and Obstetrics stages I to IIB (2018) who had undergone hysterectomy.[20] This search covered the period from each database's inception to October 2024. Additionally, we manually searched the ClinicalTrials.gov website for further relevant trials or supplementary data from identified studies. References from included articles and related systematic reviews were also examined for additional studies. Two reviewers independently selected and evaluated eligible studies, with discrepancies resolved through discussion with a third reviewer. Detailed search strategies are provided in Appendix E2.

Eligibility criteria

RCTs eligible for inclusion involved women who received diagnoses of stage I to II B cervical cancer who had undergone hysterectomy and exhibited at least one of the following risk factors: parametrial invasion, positive surgical margins, poorly differentiated tumors, deep stromal invasion, positive lymph nodes, tumor diameter >4 cm, nonsquamous histology, or lymphovascular invasion. These trials also used a minimum of 2 PATs, reported outcomes on either OS or DFS, without language restriction. Only RCTs published in peer-reviewed journals were considered, with conference abstracts and crossover trials excluded.

Screening process

We employed EndNote 20 to remove duplicate entries from the database search results. After this, 2 reviewers independently screened articles based on their titles and abstracts, excluding any unclear entries. Selected articles underwent a thorough review, with any discrepancies resolved through discussion and consensus with a third reviewer. Finally, adhering to our inclusion criteria, both reviewers conducted full-text assessments of eligible RCTs based on their titles and abstracts.

Data extraction

Two reviewers prepared tables in advance and collected data from eligible RCTs. The data covered study characteristics (authors, year, country and region, study design, cervical cancer staging, and pathological type), population details (average age, follow-up duration, and number of participants), and intervention specifics (names of interventions, methods of use, and intervention frequencies). Our primary focus was on OS and DFS, emphasizing HRs and their 95% CIs. Secondary outcomes included recurrence, local recurrence, distant metastasis, and safety events such as myelosuppression, all analyzed as binary variables. A third reviewer resolved any discrepancies.

Quality assessment of evidence

Two independent reviewers evaluated eligible RCTs using the Cochrane risk of bias tool (RoB 2.0) across 5 key areas: randomization process, adherence to planned interventions, completeness of outcome data, outcome measurement, and selection of reported results. Trials are categorized based on their bias risk.[21] Those with consistently low risk across all domains are considered reliable. If there are concerns in one domain, trials are categorized as having some concerns. Trials classified as high risk exhibit significant bias in at least one domain or multiple domains, indicating less confidence in the results. Furthermore, we employed the Confidence in Network Meta-Analysis (CINeMA) framework to comprehensively assess the certainty of evidence in NMA, considering biases within studies, reporting practices, indirectness, precision, heterogeneity, and inconsistency across 6 critical domains.[22,23] Any disagreements were resolved through consensus.

Statistical analysis

This NMA was conducted using a Bayesian framework, which facilitates the integration of direct and indirect evidence to compare and rank multiple treatments effectively. The Bayesian approach offered a robust method for estimating treatment effects and managing complex data structures. We assessed transitivity by comparing clinical and methodological variable distributions between studies that provided direct and indirect evidence.[24] We used a node-splitting technique to evaluate the alignment between direct and indirect estimates within each closed-loop. At the same time, a design treatment interaction model was used to assess the entire network.[24-26]

NMA was executed using both Stata 17.0 and R 4.0. HRs measured the effects of DFS and OS, and their 95% CIs, and myelosuppression was assessed through relative risk (RR) and their 95% CIs. When HRs and their 95% CIs were not directly reported, data extraction and transformation were performed following the methods by Tierney et al[27] and Parmar et al.[28] Direct comparisons were performed using a random-effects model to account for heterogeneity in the study. Heterogeneity was assessed using τ values, categorized based on established guidelines as low (<0.04), low-to-moderate (0.04-0.16), moderate-to-high (0.16-0.36), and high (>0.36).

A random-effects NMA was conducted within a Bayesian framework for indirect comparisons, and network diagrams were generated using Stata 17.0 to identify potential biases. A Bayesian approach in the gemtc package of R 4.0 initiated 4 Markov chains with 50,000 iterations each, discarding the first 20,000 as burn-in to eliminate initial value influences. Surface under the cumulative ranking (SUCRA) values were calculated to rank the effectiveness of treatments, with a value of 1 indicating the best and 0 for the worst treatment.

Results

Literature selection and study characteristics

The database search yielded 2867 studies, from which 950 duplicates were removed. Screening of titles and abstracts excluded another 1866 studies. After thoroughly reviewing the full 51 texts, 19 RCTs involving 2937 patients met our inclusion criteria (Fig. 1).[4,9,10,29-44] These RCTs were conducted across 4 countries, with sample sizes ranging from 52 to 1048 participants. The primary cancer type was squamous cell carcinoma, and most studies used platinum-based CT. Radiation doses were consistent across studies, while patient ages ranged from 23 to 73 years, and intervention durations varied between 24 and 60 months (Appendix E3, Table E1). Among these 19 studies, 18 reported DFS, and 17 reported OS.

Risk of bias, certainty of evidence, and consistency

Appendix E4 summarizes the risk of bias for each trial. A key limitation across studies was the lack of detailed information on blinding methods for participants, researchers, and assessors. Of the 19 trials reviewed, 18 (94.7%) demonstrated a low risk of bias in random sequence generation. Six studies (31.5%) had a low risk of bias regarding adherence to intended interventions, and all 19 studies showed a low risk in terms of missing outcome data, outcome measurement, and selection of reported results. Overall, 1 study (5.2%) was deemed high risk for bias, 12 studies (63.1%)

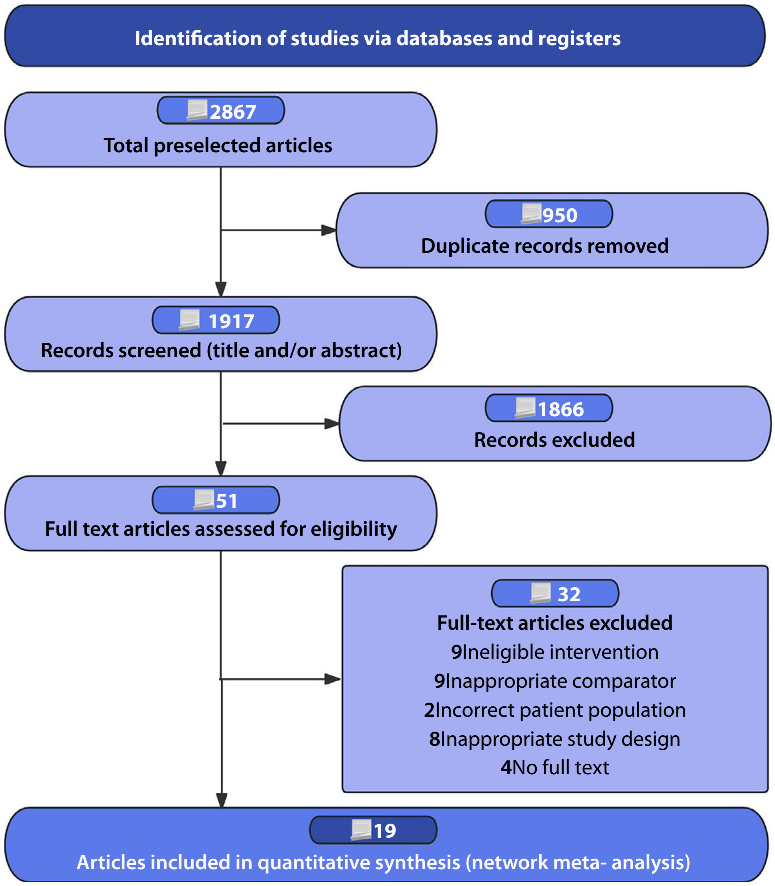

Fig. 1. Flow diagram of preferred reporting items identified, included, and excluded for systematic reviews and meta-analyses (PRISMA).

raised some concerns, and 6 studies (31.5%) had a low risk of bias.

Our evaluation of the alignment between direct and indirect evidence showed strong consistency across all comparisons. Furthermore, most outcomes did not reveal significant statistical evidence of global inconsistency. Results from the node-splitting method further supported this consistency (Appendix E5, Fig. E1). Additionally, density plots, trajectory plots, and the convergence diagnostic plot confirmed the robustness of the data (Appendix E6 and E7). The τ^2 results showed a few instances of significant heterogeneity within the network, though most comparisons exhibited low-to-moderate levels of heterogeneity (Appendix E5, Table E2). Evaluating the evidence quality with CINeMA, we found that most pairwise comparisons had low confidence, with only a few demonstrating moderate-to-high confidence (Appendix E7). All networks adhered to the transitivity principle, ensuring the validity of indirect comparisons (Appendix E7, Table E3).

Primary outcome

OS

Our NMA evaluated OS outcomes across 17 trials involving 2837 participants. Figure 2 shows both direct and indirect comparisons of various PATs, with thicker connecting lines reflecting more frequent comparisons between CCRT and RT. The forest plot suggests a trend toward improved OS with 4 alternative PATs compared with RT alone, although CT did not show a statistically significant advantage over RT (Fig. 2). Among the therapies, CCRT + CT demonstrated the most substantial improvement in OS (HR, 0.46; 95% CI, 0.28-0.71; SUCRA 95%, high-certainty evidence). CCRT alone (HR, 0.71; 95% CI, 0.5-0.95; SUCRA 48%, moderate-certainty evidence) also showed significant OS benefits. Compared with SCRT, RT was associated with worse OS (HR, 1.59; 95% CI, 1.04-2.33), with SCRT ranking moderately well in SUCRA (65%, high-certainty evidence) (Figs. 2 and 3). The league table indicated no statistically significant differences in OS between other pairs of PATs (Appendix E8, Table E4). Based on CINeMA assessments, the quality of evidence for OS was predominantly rated as low to moderate (Appendix E7, Table E5).

DFS

Our NMA assessed DFS outcomes across 18 trials, including 2857 participants. As illustrated in Figure 4, the analysis encompassed both direct and indirect comparisons of various PATs. Thicker connecting lines in the diagram indicate more frequent comparisons between CCRT and RT. The

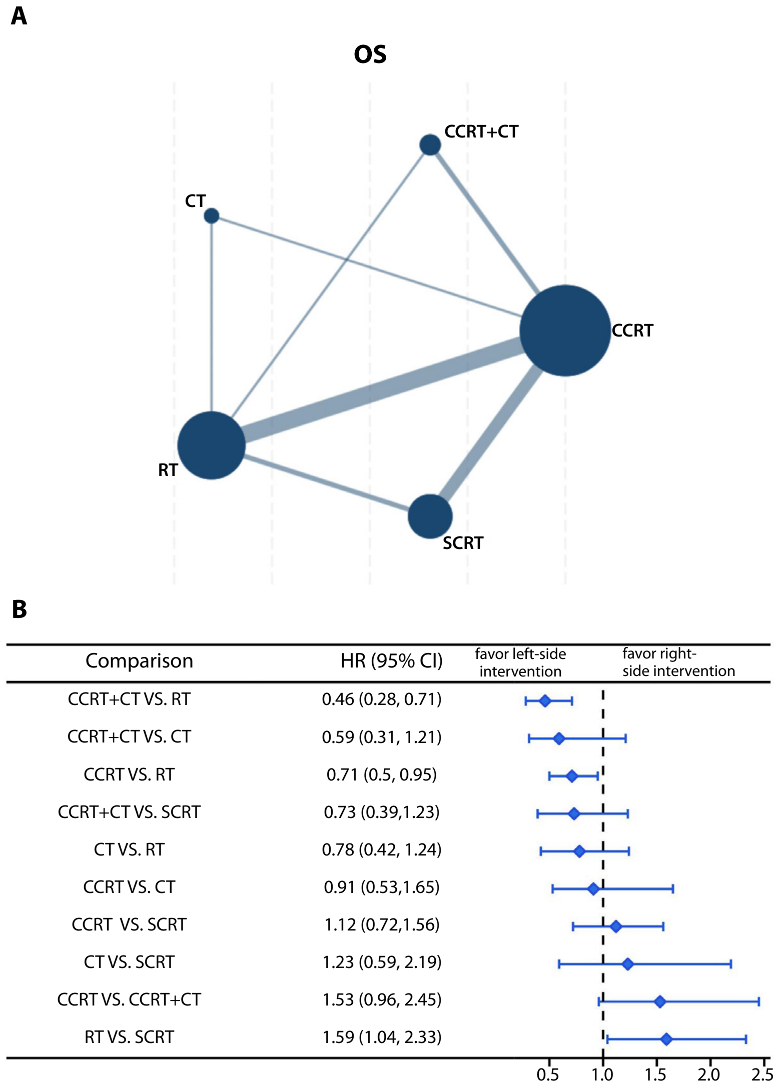

Fig. 2. Network and Forest plot of available comparisons of postoperative adjuvant therapies for overall survival (OS). (A) Network diagram illustrating the direct comparisons among the 5 postoperative adjuvant therapies: concurrent chemoradiation therapy (CCRT), radiation therapy (RT), chemotherapy (CT), sequential chemoradiation therapy (SCRT), and CCRT followed by consolidation chemotherapy (CCRT + CT). The size of each node is proportional to the total number of participants assigned to each treatment, whereas the thickness of the connecting edges reflects the number of randomized patients in each direct comparison. (B) Forest plot presenting hazard ratios (HRs) and corresponding 95% confidence intervals (CIs) derived from the network meta-analysis for OS. An HR <1 favors the treatment on the left side of the comparison. Results are ordered by intervention pairs to facilitate the interpretation of treatment effects.

forest plot suggests a trend of improved DFS with 4 alternative PATs compared with RT alone for women with cervical cancer; however, only the combination of CCRT + CT demonstrated a statistically significant advantage over RT (Fig. 4). In particular, CCRT + CT showed a significant improvement in DFS, with a HR of 0.48 (95% CI, 0.23-0.91; SUCRA 88%), although evidence quality was rated as low (Fig. 5). Additional results from the league table indicate no statistically significant differences in DFS among other pairs of PATs (Appendix E8, Table E6). Based on CINeMA evaluations, the overall quality of evidence supporting DFS improvement was generally rated as low (Appendix E7, Table E7).

Secondary outcome

Recurrence, distant recurrence, and distant metastasis

Our NMA on recurrence rates included 9 trials with 974 participants, evaluating recurrence risks across different PATs. Direct comparisons revealed significant differences in recurrence risk between RT and 3 PATs: CCRT + CT, CCRT alone, and SCRT. Among these, CCRT + CT achieved the most significant reduction in relative risk (RR, 0.45; 95% CI, 0.15-0.83; SUCRA, 85%), followed by CCRT (RR, 0.59; 95% CI, 0.24-0.97; SUCRA, 68%) and SCRT (RR, 1.17; 95% CI, 1.01-3.28; SUCRA, 40%). The league table

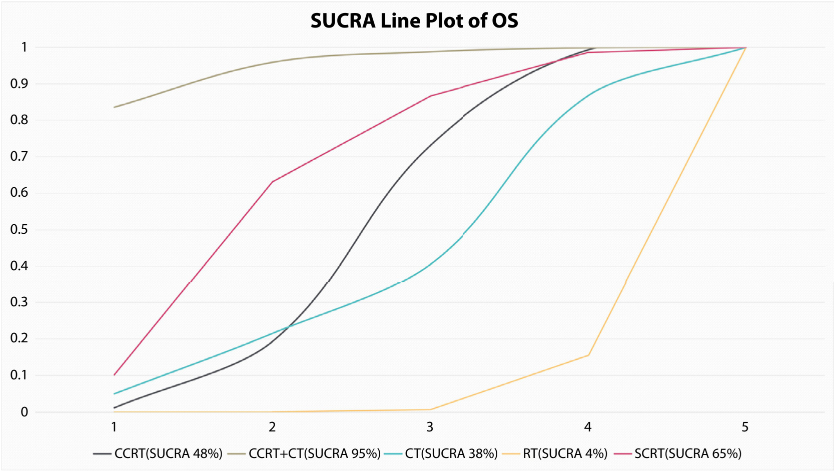

Fig. 3. The surface under the cumulative ranking (SUCRA) plot of postoperative adjuvant therapies for overall survival (OS). The SUCRA plot displays the cumulative probability of each treatment ranking at each possible position (from 1st to 5th). A larger area under the curve (closer to 1) indicates a higher probability of that treatment being among the most effective for improving OS. In this analysis, CCRT + CT had the highest SUCRA value (95%), followed by sequential chemoradiation therapy (SCRT) (65%), concurrent chemoradiation therapy (CCRT) (48%), chemotherapy (CT) (38%), and radiation therapy (RT) (4%).

indicated no statistically significant differences in recurrence risk among other PATs (Appendix E8, Table E8). Regarding distant recurrence and metastasis rates, our analysis found no statistically significant differences among the PATs. However, the SUCRA rankings suggested that CCRT + CT and CCRT may show potential in reducing these outcomes. Detailed comparisons and rankings can be found in the SUCRA data (Appendix E9, Figs. E2-E4) and supplementary tables (Appendix E8, Tables E8-E10).

Adverse events

To evaluate adverse events—including myelosuppression, proctitis, cystitis, and nausea/vomiting—we analyzed data from 13 trials, 8 studies, and 5 studies, respectively. The analysis found no significant differences in the risk of these events across the various PATs. However, SUCRA rankings consistently placed RT as the top option for minimizing each of these adverse events, suggesting a potential advantage in reducing their occurrence. Detailed SUCRA data and comparisons of these adverse events are available in Appendix E9 (Figs. E5-E8) and Appendix E8 (Tables E11-E14).

Sensitivity analyses and meta-regressions

To ensure the robustness of our findings, we conducted sensitivity analyses using HRs estimated through multiple methods and by excluding lower-quality studies. This approach helped verify the consistency of outcome measures when different estimation methods or study qualities were applied. Sensitivity analyses on HRs for OS and DFS, using CCRT as the reference, confirmed the stability of our results. Excluding high-risk studies did not alter the findings, indicating low sensitivity and reliable outcomes, as shown in Appendix E10. Additionally, a meta-regression analysis was performed to assess the impact of potential baseline effect modifiers, including staging, pathology classification, and mean age, on the primary outcomes. None of these factors significantly influenced the results (Appendix E11).

Discussion

Treatment strategies for cervical cancer continue to evolve, with growing emphasis on optimizing surgical techniques and tailoring approaches to individual patient characteristics. For patients with stage I to IIB disease, the standard options include surgery or RT. When fertility preservation is desired, procedures such as cervical conization or trachelectomy may be appropriate[45]; otherwise, hysterectomy remains the most widely used approach. Among surgical methods, both abdominal and laparoscopic techniques are common, with laparoscopy often preferred due to its advantages in reducing blood loss, accelerating recovery, and lowering complication rates.[46,47] However, evidence has emerged questioning the oncologic safety of minimally invasive surgery in certain patients. A 2020 meta-analysis by Nitecki et al[48] reported higher recurrence and mortality rates associated with minimally invasive radical hysterectomy compared with open surgery in early-stage disease. These concerns were confirmed by the Laparoscopic Approach to Cervical Cancer RCT, which found that minimally invasive surgery was associated with significantly worse DFS and OS in patients with stage IA2 and IB1 cervical cancer.[49] As a result, current guidelines now place greater emphasis on aligning surgical choice with tumor

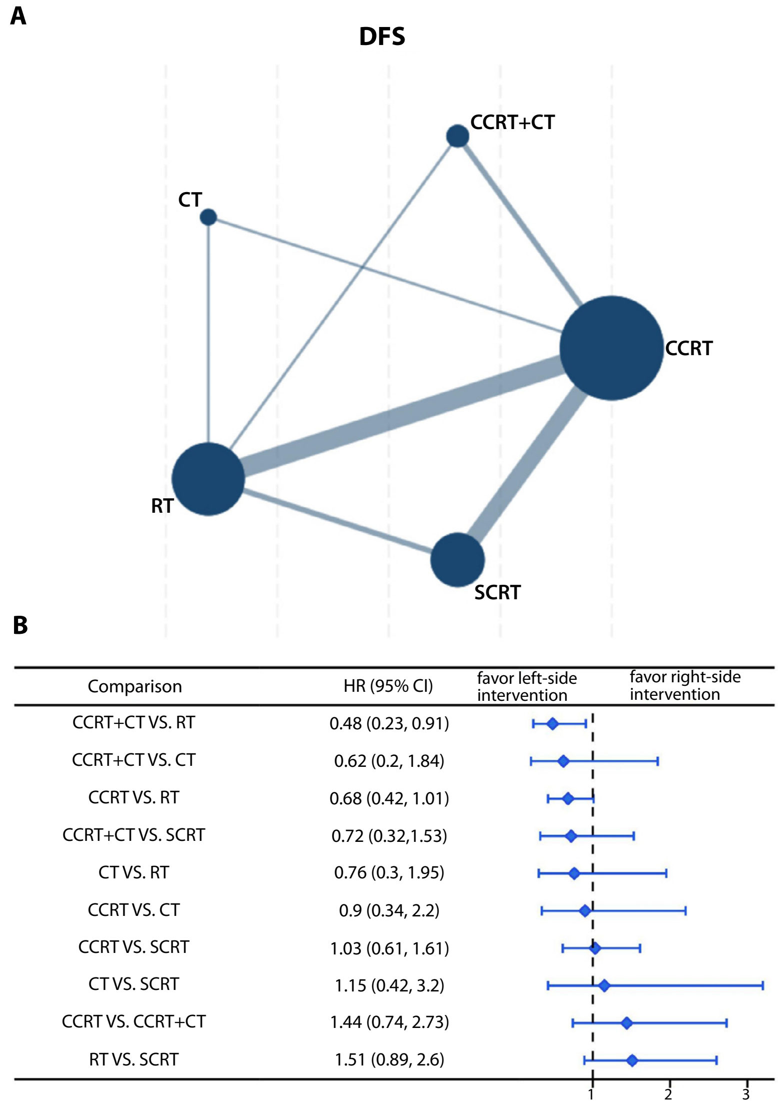

Fig. 4. Network and Forest plot of available comparisons of postoperative adjuvant therapies for disease-free survival (DFS). (A) Network diagram displaying available head-to-head comparisons among 5 postoperative adjuvant therapies: concurrent chemoradiation therapy (CCRT), radiation therapy (RT), chemotherapy (CT), sequential chemoradiation therapy (SCRT), and CCRT followed by consolidation chemotherapy (CCRT + CT). The size of each node represents the total number of patients receiving each intervention, and the width of the connecting lines reflects the number of direct comparisons between treatments. (B) Forest plot presenting hazard ratios (HRs) and 95% confidence intervals (CIs) from the network meta-analysis for overall survival. HRs <1 indicate a survival benefit in favor of the left-side intervention. Comparisons are sorted by intervention pairs for easier interpretation.

size, disease stage, and other risk factors. More recently, the SHAPE trial further refined surgical decision-making by demonstrating that, in patients with low-risk IB1 tumors (defined as ≤2 cm with limited stromal invasion), simple hysterectomy was noninferior to radical hysterectomy in terms of pelvic recurrence. Importantly, simple hysterectomy was also associated with fewer postoperative urinary complications, supporting a less aggressive surgical approach in carefully selected patients.[50]

Postoperative treatment decisions in cervical cancer are largely determined by pathological risk profiles. However, the most effective adjuvant approach remains a subject of ongoing debate. For patients with postoperative risk factors, CCRT is frequently administered to reduce recurrence rates.[51] This approach combines CT with RT to enhance radiosensitivity, potentially achieving better treatment outcomes and limiting tumor cell repair associated with RT alone. Currently, there is no established standard protocol for early-stage cervical cancer, and the optimal chemoradiation therapy regimen remains a topic of debate. Although chemoradiation therapy has been shown to improve survival, it can also cause adverse effects, such as gastrointestinal symptoms and myelosuppression, which may influence recovery, treatment duration, and OS benefits. Previous studies in this area have primarily been retrospective and based on small sample sizes. By employing NMA, this study

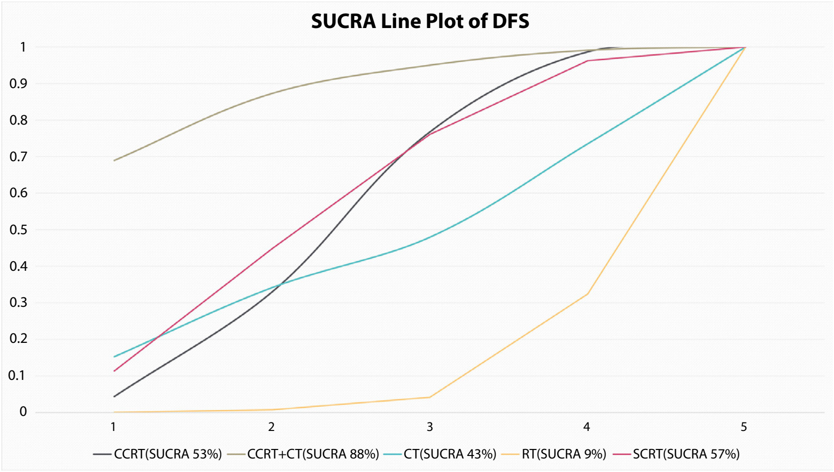

Fig. 5. The surface under the cumulative ranking (SUCRA) plot of postoperative adjuvant therapies for disease-free survival (DFS). The SUCRA plot summarizes the cumulative probability that each postoperative adjuvant therapy ranks among the most effective for improving DFS. Each curve represents one treatment's cumulative probability of ranking from best to worst. A higher SUCRA value (closer to 1) indicates a higher likelihood of being among the top-performing treatments. In this analysis, CCRT + CT had the highest SUCRA score (88%), followed by sequential chemoradiation therapy (SCRT) (57%), concurrent chemoradiation therapy (CCRT) (53%), chemotherapy (CT) (43%), and radiation therapy (RT) (9%).

aimed to integrate the existing evidence to evaluate the various PATs available comprehensively.

Meanwhile, advances in the treatment of unresected, locally advanced cervical cancer have led to new strategies that, although impactful, are not directly applicable to the postoperative setting. Recently, 2 pivotal RCTs—INTERLACE[52] and KEYNOTE-A18[53]—have introduced significant advances in the treatment of locally advanced cervical cancer. INTERLACE demonstrated the survival benefit of induction CT before standard CRT, whereas KEYNOTE-A18 showed that combining immunotherapy with CRT and maintenance therapy improved both progression - free survival and OS. However, both trials focused on patients with unresected, locally advanced disease (International Federation of Gynecology and Obstetrics stages IB2-IVA), rather than on early-stage patients undergoing radical surgery. As such, their findings—although highly relevant for the evolving landscape of primary treatment—do not directly address the postoperative adjuvant context evaluated in our study.

Principal findings

This NMA comprehensively evaluated the efficacy and safety of 5 different PATs in patients with cervical cancer. The primary PATs included CCRT + CT, CCRT, CT, RT, and SCRT. We assessed OS, DFS, recurrence-related outcomes, and treatment-related adverse events across 19 eligible RCTs involving 2937 participants. Our data demonstrated that CCRT + CT was the most effective PAT for improving OS with high confidence and DFS with low confidence. It was also the most effective PAT for reducing recurrence rates. Additionally, both CCRT and SCRT showed promising results in enhancing OS and DFS and reducing recurrence rates. However, whether the various PATs differ in their impact on distant recurrence, distant metastasis, myelosuppression, proctitis, cystitis, or nausea and vomiting remained uncertain.

Comparisons with other studies

Regarding OS improvement, CCRT showed a statistically significant advantage over RT (HR, 0.71; 95% CI, 0.5-0.95), with CCRT yielding better OS outcomes, consistent with prior studies and guideline recommendations.[16,54] Similarly, SCRT demonstrated a significant OS improvement compared with RT (HR, 1.59; 95% CI, 1.04-2.33), outperforming RT regarding OS. Although the difference between SCRT and CCRT was not statistically significant, the SUCRA ranking suggested that SCRT might be a preferable option for improving OS (SUCRA = 65%). Four studies have directly compared SCRT and CCRT. Although Huang et al[10] found a statistically significant advantage for SCRT over CCRT, the other studies reported no significant differences between the 2 therapies. These studies' relatively small sample sizes may contribute to potential false-negative results. Thus, further high-quality, large-sample RCTs are needed to better evaluate the comparative efficacy of SCRT and CCRT. Our analysis also found no statistically significant difference between CCRT and CT, contrasting with Zhang et al's[11] meta-analysis. This discrepancy may be due to differences in study populations, as Zhang et al's[11] analysis included many patients with stage IIB cervical cancer and a small proportion of IIIA or IIIB patients. Additionally, a retrospective study by Zhong et al[55] indicated that among patients with more than 3 positive lymph nodes, lymphovascular space invasion, or deep stromal invasion beyond one-third, CCRT + CT improved OS more effectively than CCRT alone. Although our study did not find a statistically significant difference between CCRT + CT and CCRT, SUCRA rankings suggested a potential OS benefit with CCRT + CT over CCRT.

CCRT + CT demonstrated a statistically significant improvement in DFS compared with RT (HR, 0.48; 95% CI, 0.23-0.91), suggesting CCRT + CT as a more effective option. SUCRA rankings supported this, placing CCRT + CT as the top choice for enhancing DFS (SUCRA score = 88%), in line with Ao et al.[56] DFS and OS SUCRA rankings positioned CCRT + CT first, with CCRT in third place. A retrospective study by Zhong et al[55] also noted that CCRT + CT improved DFS over CCRT among patients with more than 3 positive lymph nodes, lymphovascular invasion, or deep stromal invasion. However, CCRT + CT was also linked to higher rates of grade 3/4 myelosuppression, which is consistent with our findings.

Our analysis showed that CCRT + CT, CCRT, and SCRT were more effective than RT in reducing recurrence, though studies revealed no significant differences in distant recurrence or metastasis. Tursun et al[57] similarly observed no significant differences in distant recurrence or metastasis between CCRT and RT, which supports our findings regarding these endpoints. Separately, Xia et al[58] reported no significant difference in recurrence rates between CT and CCRT after hysterectomy, suggesting that the benefit of adding CT may depend on treatment combinations and patient profiles.

Regarding myelosuppression and other adverse events, no statistically significant differences were observed across all PATs; however, SUCRA rankings suggested that RT may be the most favorable for minimizing adverse events. Only 13 studies reported on myelosuppression, and inconsistencies in reporting prevented data pooling for meta-analysis. Many studies reported different types of myelosuppression separately—such as anemia, leukopenia, thrombocytopenia, and neutropenia—making it difficult to determine overall incidence. Future research should report myelosuppression in more detail, because it is a common complication associated with chemoradiation therapy—particularly due to CT—and may affect treatment duration and overall therapeutic efficacy.[59]

Policy implications

This study underscores the need for updated treatment guidelines for early-stage cervical cancer, particularly in considering CCRT + CT. Given its potential benefits in improving OS and DFS and reducing recurrence rates, CCRT + CT may represent a promising option for patients at high risk of recurrence or with adverse pathological features. Future guideline committees could evaluate the evidence presented here to determine whether incorporating CCRT + CT into treatment protocols is justified, especially for selected subgroups such as those with multiple positive lymph nodes or deep stromal invasion.

Our findings also emphasize the importance of managing the adverse effects of CCRT + CT, notably grade 3/4 myelosuppression. Health systems should prioritize supportive services, including close monitoring for myelosuppression, to mitigate risks and support continuous care. Developing supportive care guidelines for managing hematologic and gastrointestinal complications can help patients adhere to treatment, enhancing the overall benefits of CCRT + CT. Furthermore, investing in training and resources for adverse event management could improve both immediate and long-term outcomes for patients undergoing intensive adjuvant therapies.

This study also highlights the need for further large-scale RCTs to confirm the relative effectiveness of different PATs. Policymakers should support and fund high-quality research to address these evidence gaps, because current data are often limited by small sample sizes or retrospective designs. We also recommend standardizing criteria for reporting adverse events such as myelosuppression, because inconsistent reporting across studies impedes comprehensive analysis. Clearer reporting standards would allow for more reliable meta-analyses and support the development of precise, patient-centered treatment protocols.

By aligning treatment guidelines with current evidence and investing in ongoing research, health policymakers can help optimize outcomes for cervical cancer patients, addressing both survival rates and quality of life.

Strengths and limitations of this study

This study offers a comprehensive assessment of adjuvant therapies for cervical cancer, marking the first NMA to evaluate both efficacy and safety outcomes for these treatments. The included studies featured patients with consistent characteristics, including similar age ranges, stage I to IIB disease, a predominance of squamous cell carcinoma, and comparable radiation and CT doses—mostly platinum-based—which strengthened cross-study comparability. Using HRs to capture survival outcomes and duration, this analysis provides a robust estimation of treatment effects, unlike previous studies that relied on RR or odds ratios. The NMA also enabled direct comparisons among 5 treatment strategies, helping to identify the optimal approach and offering practical guidance for clinical decision-making.

This study also has limitations. Many included studies did not report allocation concealment methods, which may introduce selection bias. The analysis also primarily included studies conducted in China, with limited representation from other countries, potentially leading to geographic bias. Despite this, our comprehensive search was conducted without language restrictions, and all findings were validated through inconsistency analyses, CINeMA assessments, and sensitivity analyses to ensure robustness. Additionally, studies included both laparoscopic and open abdominal hysterectomies, and the potential impact of these surgical methods on survival and adverse events remains debated, which may have influenced the results. Finally, our safety analysis was restricted to 4 adverse outcomes—

myelosuppression, cystitis, proctitis, and nausea/vomiting—due to inconsistent reporting of other events, limiting broader insights into treatment-related complications. Another limitation of this study is the variability in how “high-risk” patients were defined across the included trials, which encompassed a wide range of pathological features. Because our analysis relied on aggregated study-level data, we could not examine outcomes based on the number or specific combinations of risk factors. This reliance on aggregate data also limited our ability to account for patient-level heterogeneity, treatment compliance, and interaction effects between covariates. As a result, subgroup-specific findings—particularly for high-risk populations—should be interpreted with caution. This constraint limited our ability to assess the effectiveness of adjuvant therapies in distinct patient subgroups. To overcome these limitations, future research should consider individual patient data meta-analyses, which would enable more nuanced stratification and facilitate personalized treatment recommendations.

Conclusions

This study offers a comprehensive assessment of PATs for early-stage cervical cancer. In our pooled analysis, CCRT + CT demonstrated a relatively favorable trend in improving OS, DFS, and reducing recurrence, suggesting potential benefit for certain subgroups of patients with high-risk features. However, the association of CCRT + CT with increased risks of myelosuppression highlights the need for enhanced management of adverse effects to ensure treatment adherence and optimize outcomes. It is important to note that the definition of “high-risk” varied across the included studies, and our analysis was based on aggregate-level data without access to individual patient-level information. As a result, we were unable to further stratify patients based on the number or combinations of specific risk factors. Although our findings highlight the potential clinical value of CCRT + CT in selected high-risk populations, further large-scale, high-quality RCTs are warranted to validate these results and to refine adjuvant treatment recommendations.

References

1. World Health Organization. Human papillomavirus (HPV) and cervical cancer. 2021. Accessed November 18, 2021. http://www.who.int/mediacentre/factsheets/fs380/en/.
2. International Agency for Research on Cancer. Global Cancer Observatory; *Cancer today* 2022. Accessed May 23, 2025; https://gco.iarc.who.int/media/globocan/factsheets/cancers/23-cervix-uteri-fact-sheet.pdf.
3. National Comprehensive Cancer Network (NCCN). NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®) for Cervical Cancer. version 4.2025. Accessed August 28, 2025. https://www.nccn.org/professionals/physician_gls/pdf/cervical.pdf.
4. Peters III WA, Liu PY, Barrett II RJ, et al. Concurrent chemotherapy and pelvic radiation therapy compared with pelvic radiation therapy alone as adjuvant therapy after radical surgery in high-risk early-stage cancer of the cervix. *J Clin Oncol* 2000;18:1606-1613.
5. Rotman M, Sedlis A, Piedmonte MR, et al. A phase III randomized trial of postoperative pelvic irradiation in stage IB cervical carcinoma with poor prognostic features: Follow-up of a gynecologic oncology group study. *Int J Radiat Oncol Biol Phys* 2006;65:169-176.
6. Sedlis A, Bundy BN, Rotman MZ, Lentz SS, Muderspach LI, Zaino RJ. A randomized trial of pelvic radiation therapy versus no further therapy in selected patients with stage IB carcinoma of the cervix after radical hysterectomy and pelvic lymphadenectomy: A Gynecologic Oncology Group Study. *Gynecol Oncol* 1999;73:177-183.
7. Marth C, Landoni F, Mahner S, et al. Cervical cancer: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. *Ann Oncol* 2017;28(suppl 4):iv72-iv83 Published correction appears in iv262. *Ann Oncol*. 2018;29(suppl 4):iv262.
8. Sha HL, He AQ. Observation of the effects of TP regimen chemotherapy followed by radiotherapy on patients with high-risk factors after cervical cancer surgery. *J Community Med* 2016;30:622-625.
9. Sehouli J, Runnebaum IB, Fotopoulou C, et al. A randomized phase III adjuvant study in high-risk cervical cancer: simultaneous radiochemotherapy with cisplatin (S-RC) versus systemic paclitaxel and carboplatin followed by percutaneous radiation (PC-R): A NOGGO-AGO Intergroup Study. *Ann Oncol* 2012;23:2259-2264.
10. Huang H, Feng YL, Wan T, et al. Effectiveness of sequential chemoradiation vs concurrent chemoradiation or radiation alone in adjuvant treatment after hysterectomy for cervical cancer: The STARS phase 3 randomized clinical trial. *JAMA Oncol* 2021;7:361-369.
11. Zhang YF, Fan Y, Zhang P, Ruan JY, Mu Y, Li JK. Cervical cancer recurrence and patient survival after radical hysterectomy followed by either adjuvant chemotherapy or adjuvant radiotherapy with optional concurrent chemotherapy: A systematic review and meta-analysis. *Front Oncol* 2022;12:823064.
12. Matsuo K, Shimada M, Aoki Y, et al. Comparison of adjuvant therapy for node-positive clinical stage IB-IIB cervical cancer: Systemic chemotherapy versus pelvic irradiation. *Int J Cancer* 2017;141:1042-1051.
13. Liu H, Ma X, Sun C, et al. Concurrent chemoradiotherapy followed by adjuvant chemotherapy versus concurrent chemoradiotherapy alone in locally advanced cervical cancer: A systematic review and meta-analysis. *Front Oncol* 2022;12:997030.
14. Marchand G, Masoud AT, Abdelsattar A, et al. Meta-analysis of laparoscopic radical hysterectomy, excluding robotic assisted versus open radical hysterectomy for early stage cervical cancer. *Sci Rep* 2023;13:273.
15. Ronsini C, De Franciscis P, Carotenuto RM, Pasanisi F, Cobellis L, Colacurci N. The oncological implication of sentinel lymph node in early cervical cancer: A meta-analysis of oncological outcomes and type of recurrences. *Medicina (Kaunas)* 2022;58:1539.
16. Guo Q, Wang R, Jin D, et al. Comparison of adjuvant chemoradiotherapy versus radiotherapy in early-stage cervical cancer patients with intermediate-risk factors: A systematic review and meta-analysis. *Taiwan J Obstet Gynecol* 2022;61:15-23.
17. Zhou ZR, Zhang TS, Li B, et al. Extraction and transformation of appropriate data for meta-analysis in survival curves. *Chin J Evid Based Cardiovasc Med* 2014;6:243-247.
18. Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *BMJ* 2021;372:n71.
19. Hutton B, Salanti G, Caldwell DM, et al. The PRISMA extension statement for reporting of systematic reviews incorporating network meta-analyses of health care interventions: Checklist and explanations. *Ann Intern Med* 2015;162:777-784.
20. Bhatla N, Aoki D, Sharma DN, Sankaranarayanan R. Cancer of the cervix uteri. *Int J Gynecol Obstet* 2018;143(suppl 2):22-36 Published correction appears in *Int J Gynecol Obstet*. 2024;164:1229-1230.
21. Sterne JAC, Savović J, Page MJ, et al. RoB 2: A revised tool for assessing risk of bias in randomised trials. *BMJ* 2019;366:l4898.
22. Nikolakopoulou A, Higgins JPT, Papakonstantinou T, et al. CINeMA: An approach for assessing confidence in the results of a network meta-analysis. *PLoS Med* 2020;17:el003082.
23. Papakonstantinou T, Nikolakopoulou A, Higgins T, Egger M, Salanti G. CINeMA: Software for semiautomated assessment of the confidence in the results of network meta-analysis. *Campbell Syst Rev* 2020;16:e1080.

24. Furukawa TA, Salanti G, Atkinson LZ, et al. Comparative efficacy and acceptability of first-generation and second-generation antidepressants in the acute treatment of major depression: Protocol for a network meta-analysis. *BMJ Open* 2016;6:e010919.
25. Dias S, Welton NJ, Sutton AJ, Caldwell DM, Lu G, Ades AE. Evidence synthesis for decision making 4: Inconsistency in networks of evidence based on randomized controlled trials. *Med Decis Making* 2013;33:641-656.
26. Higgins JP, Jackson D, Barrett JK, Lu G, Ades AE, White IR. Consistency and inconsistency in network meta-analysis: Concepts and models for multi-arm studies. *Res Synth Methods* 2012;3:98-110.
27. Tierney JF, Stewart LA, Ghersi D, Burdett S, Sydes MR. Practical methods for incorporating summary time-to-event data into meta-analysis. *Trials* 2007;8:16.
28. Parmar MK, Torri V, Stewart L. Extracting summary statistics to perform meta-analyses of the published literature for survival endpoints. *Stat Med* 1998;17:2815-2834 Published correction appears in *Stat Med*. 2004;23:1817.
29. Curtin JP, Hoskins WJ, Venkatraman ES, et al. Adjuvant chemotherapy versus chemotherapy plus pelvic irradiation for high-risk cervical cancer patients after radical hysterectomy and pelvic lymphadenectomy (RH-PLND): A randomized phase III trial. *Gynecol Oncol* 1996;61:3-10.
30. Lahousen M, Haas J, Pickel H, et al. Chemotherapy versus radiotherapy versus observation for high-risk cervical carcinoma after radical hysterectomy: A randomized, prospective, multicenter trial. *Gynecol Oncol* 1999;73:196-201.
31. Tattersall MH, Ramirez C, Coppleson M. A randomized trial of adjuvant chemotherapy after radical hysterectomy in stage Ib-IIa cervical cancer patients with pelvic lymph node metastases. *Gynecol Oncol* 1992;46:176-181.
32. Zhao H, Li L, Su H, et al. Concurrent paclitaxel/cisplatin chemoradiotherapy with or without consolidation chemotherapy in high-risk early-stage cervical cancer patients following radical hysterectomy: Preliminary results of a phase III randomized study. *Oncotarget* 2016;7:70969-70978.
33. Lin GS, Cheng HH, Li DS. Comparative study of adjuvant concurrent chemoradiotherapy and sequential chemoradiotherapy for early-stage high-risk cervical cancer after surgery. *J Clin Oncol* 2009;14:541-543.
34. Cui YL, Wang B, Zhang JH, et al. Comparative analysis of the effects of adjuvant concurrent chemoradiotherapy versus sequential chemoradiotherapy after radical surgery for cervical cancer; *J Transl Med Electron* 2015;2:58-59. https://www.CNKI:SUN:ZHDZ.0.2015-09-033.
35. Zhen C, Wang ZH, Wang C, et al. Comparative observation of clinical efficacy between sequential chemoradiotherapy and concurrent chemoradiotherapy in early cervical cancer after surgery. *J Basic Clin Oncol* 2021;34:322-324. https://doi.org/10.3969/j.issn.1673-5412.2021.04.012.
36. Qiao ZA. Clinical analysis of adjuvant concurrent chemoradiotherapy after radical surgery for cervical cancer. *Chin J Cancer Prev Treat* 2016;23(S1):186-187. https://doi.org/10.16073/j.cnki.cjcpt.2016.s1.094.
37. Xing MH, Ge W, Ran RZ, et al. Clinical exploration of replacing standard pelvic field radiation with small pelvic field chemoradiotherapy for stage Ia2-IIa cervical cancer after surgery. *Mod Oncol Med* 2018;26:2925-2929. https://doi.org/10.3969/j.issn.1672-4992.2018.18.024.
38. Ni TT, Wu L, Chen YP, et al. Clinical observation of adjuvant chemotherapy for lymph node metastasis after concurrent chemoradiotherapy for cervical cancer after surgery. *China Med Innov* 2020;17:5-10. https://doi.org/10.3969/j.issn.1674-4985.2020.29.002.
39. Chen J. Observation of the effect of paclitaxel combined with cisplatin on lower limb lymphedema after cervical cancer surgery; *J Chronic Dis* 2018;19:223-225. https://www.CNKI:SUN:YYSO.0.2018-02-039.
40. Tian XY, Li RG, Zeng YB. Comparative study of concurrent chemoradiotherapy with intensity-modulated radiotherapy versus sequential chemotherapy after surgery for high-risk cervical cancer. *J Guangdong Med Univ* 2018;36:460-462. https://doi.org/10.3969/j.issn.1005-4057.2018.04.029.
41. Huang DN, Li GS, Huang HX, et al. Clinical analysis of concurrent chemoradiotherapy in early-stage high-risk cervical cancer after surgery. *Youjiang Med* 2007;1:20-21. https://doi.org/10.3969/j.issn.1003-1383.2007.01.010.
42. Liu W, Yu LN. Clinical observation of adjuvant chemotherapy versus radiotherapy in patients with intermediate-risk early to mid-stage cervical cancer after radical surgery. *Chin Foreign Womens Health Res* 2017;1:150-151. https://doi.org/10.3969/j.issn.2096-0417.2017.01.109.
43. Feng H. Analysis of efficacy and complications of different treatment methods for early-stage cervical cancer patients with high-risk recurrence factors after surgery; *J Pract Med Forum* 2019;23:1599-1600. https://www.CNKI:SUN:YXLT.0.2019-11-082.
44. Yang ZY. Value of chemoradiotherapy for pelvic lymph node metastasis after cervical cancer surgery; *J Clin Oncol* 2000;1:39-40. https://www.CNKI:SUN:LCZL.0.2000-01-019.
45. Alur-Gupta S, Fruchtman H, Paroder V. Fertility-sparing options for cancer patients. *Abdom Radiol (NY)* 2023;48:1618-1628.
46. Jiang F, Xiang Y. Current status and future of minimally invasive surgery for early-stage cervical cancer. *Chin J Clin Obstet Gynecol* 2021;2:113-115.
47. Kondo E, Yoshida K, Kubo-Kaneda M, et al. Does vaginal cuff creation and avoidance of a uterine manipulator improve the prognosis of total laparoscopic radical hysterectomy for early cervical cancer? A retrospective multicenter study. *Cancers (Basel)* 2022;14:4389.
48. Nitecki R, Ramirez PT, Frumovitz M, et al. Survival after minimally invasive vs open radical hysterectomy for early-stage cervical cancer: A systematic review and meta-analysis. *JAMA Oncol* 2020;6:1019-1027.
49. Ramirez PT, Frumovitz M, Pareja R, et al. Minimally Invasive versus Abdominal radical hysterectomy for Cervical Cancer. *N Engl J Med* 2018;379:1895-1904.
50. Plante M, Kwon JS, Ferguson S, et al. Simple versus radical hysterectomy in women with low-risk cervical cancer. *N Engl J Med* 2024;390:819-829.
51. Delgado G, Bundy B, Zaino R, Sevin BU, Creasman WT, Major F. Prospective surgical-pathological study of disease-free interval in patients with stage IB squamous cell carcinoma of the cervix: A Gynecologic Oncology Group study. *Gynecol Oncol* 1990;38:352-357.
52. McCormack M, Eminowicz G, Gallardo D, et al. Induction chemotherapy followed by standard chemoradiotherapy versus standard chemoradiotherapy alone in patients with locally advanced cervical cancer (GCIG INTERLACE): An international, multicentre, randomised phase 3 trial. *Lancet* 2024;404:1525-1535 Published correction appears in *Lancet*. 2025;405:468.
53. Lorusso D, Xiang Y, Hasegawa K, et al. Pembrolizumab or placebo with chemoradiotherapy followed by pembrolizumab or placebo for newly diagnosed, high-risk, locally advanced cervical cancer (ENGOT-cx11/GOG-3047/KEYNOTE-A18): A randomised, double-blind, phase 3 clinical trial. *Lancet* 2024;403:1341-1350.
54. Liu ZB, Li SY, Zhao YN, et al. Meta-analysis of concurrent chemoradiotherapy versus radiotherapy alone in postoperative patients with intermediate-risk factors for stage I-II cervical cancer. *Chin J Geriatr* 2020;40:4301-4305. https://doi.org/10.3969/j.issn.1005-9202.2020.20.014.
55. Zhong ML, Wang YN, Liang MR, Liu H, Zeng SY. Consolidation chemotherapy in early-stage cervical cancer patients with lymph node metastasis after radical hysterectomy. *Int J Gynecol Cancer* 2020;30:602-606. https://doi.org/10.13283/j.cnki.xdfckjz.2015.07.016.
56. Ao MH, Li LY. Meta-analysis of the effect of consolidation chemotherapy on the prognosis of cervical cancer after concurrent chemoradiotherapy. *Progr Mod Obstet Gynecol* 2015;24:528-530.
57. Tursun R, Alifu M, Wang YN, et al. Impact of different treatment modalities on the prognosis of stage Ib-IIa cervical squamous carcinoma patients with only intermediate-risk factors after surgery. *Xinjiang Med* 2018;48:1063-1067.
58. Xia XP, Shen W, Ni J, et al. Analysis of efficacy and quality of life of different adjuvant treatments in postoperative cervical cancer patients with intermediate-risk factors. *J Chongqing Med Univ* 2021;46:691-695. https://doi.org/10.13406/j.cnki.cyxb.002820.
59. Vitzthum LK, Heide ES, Park H, et al. Comparison of hematologic toxicity and bone marrow compensatory response in head and neck vs. cervical cancer patients undergoing chemoradiotherapy. *Front Oncol* 2020;10:1179.